Multiple testing considerations

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Course logistics:

Today: FDR, BH procedure

Friday: interim analysis, course evals

Monday: interim analysis

Wednesday: course wrap-up

- HW 10 (due Friday) is final homework
- Review questions for final will be posted

Some methods to control FWER

Suppose we test m null hypotheses $H_{0,1}, ..., H_{0,m}$. Let p_i be the corresponding p-value for test i.

Bonferroni method: Reject $H_{0,i}$ if $p_i < \alpha/m$

Holm method:

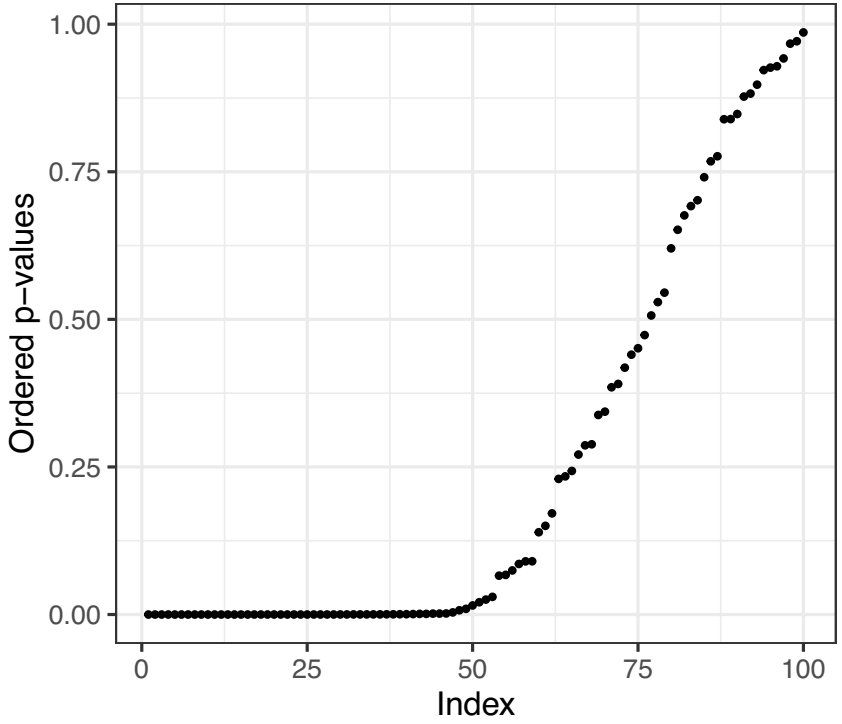
- Order the p-values $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}$
- Let $i^* = \min\left\{i : p_{(i)} > \frac{\alpha}{m-i+1}\right\}$
- Reject $H_{0,(i)}$ for all $i < i^*$

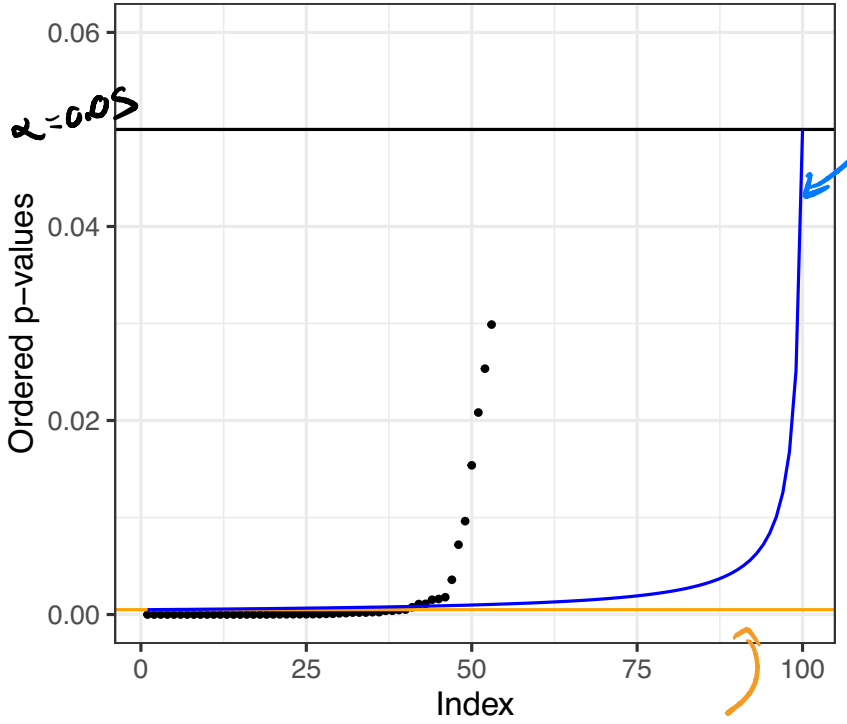
Both Holm & Bonferroni <u>always</u> control FWER

Both tend to be conservative

Holm dominates Bonferroni (always at least as powerful as Bonferroni)

Ordered p-value plots





$\frac{\alpha}{m-i+1}$

Holm's

$\alpha = 0.05$

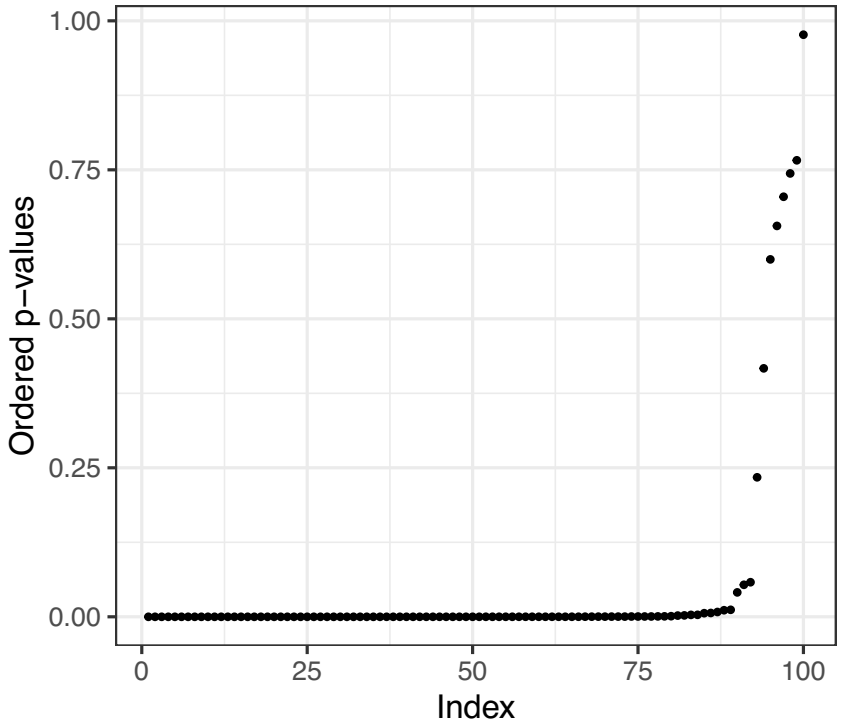
Bonferroni

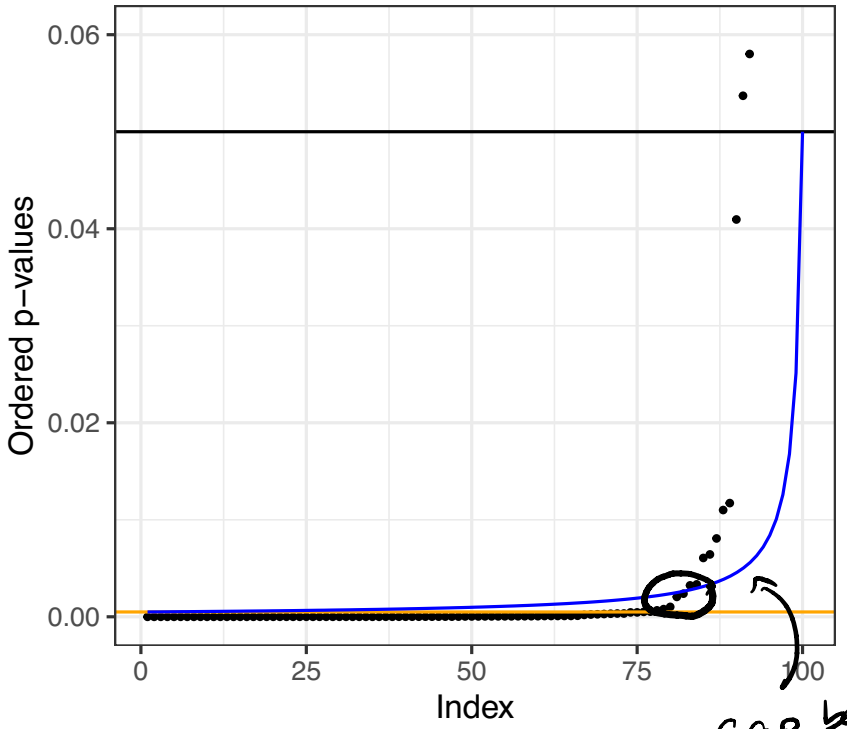
$\left(\frac{\alpha}{m}\right)$

i.e. $\frac{0.05}{100}$

Ordered p-value plots

Decreasing m_0: (now most of our hypotheses should be rejected)





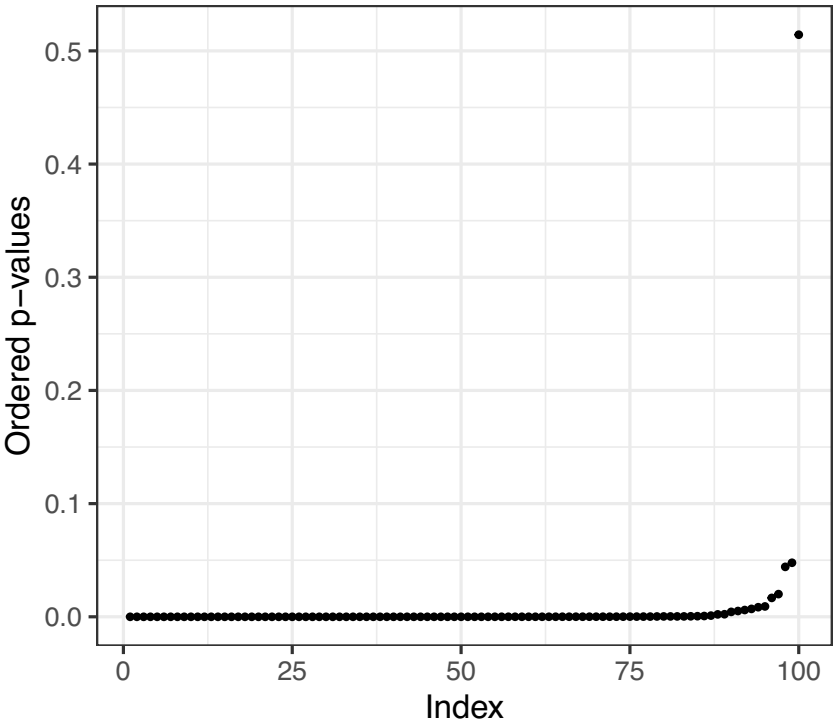
gap between Holm & Bonferroni

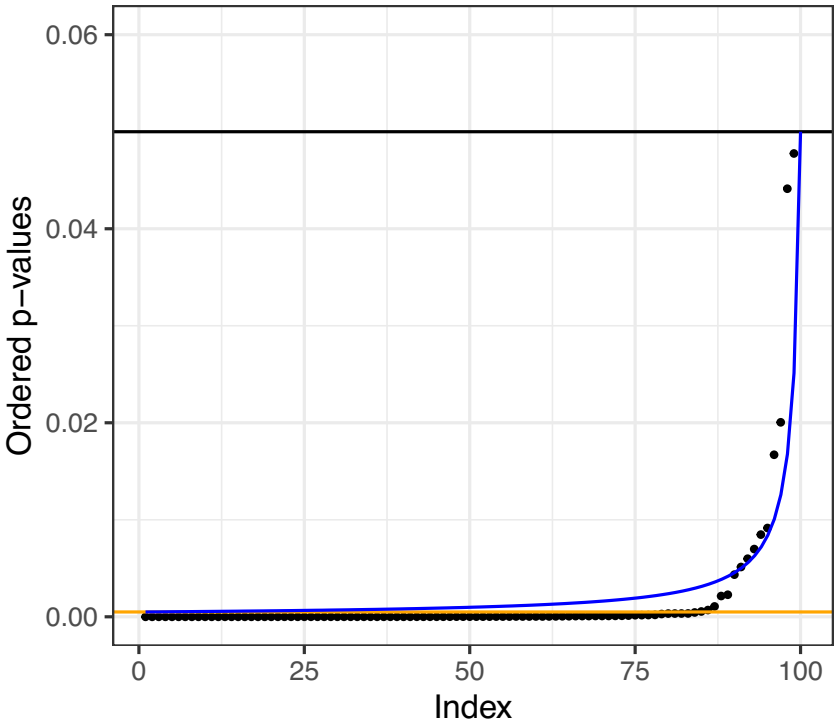
Ordered p-value plots

Holm: at least as powerful as Bonferroni

relative power increases as we decrease m_0

Decreasing m_0:





Outcomes for multiple hypothesis tests

Test m hypotheses, m_0 are truly null

	H_0 is true	H_0 is false	
Reject	V		R
Fail to reject			
	m_0		m

$$\text{FWER} = P(V > 0)$$

$$\text{FDP} = \begin{cases} \dfrac{V}{R} & \text{if } R > 0 \\ 0 & \text{if } R = 0 \end{cases}$$

(false discovery proportion)

$$\text{FDR} = \mathbb{E}[\text{FDP}]$$

(false discovery rate)

False discovery rate

controlling FWER $\Rightarrow$ controlling FDR

controlling FDR $\not\Rightarrow$ controlling FWER

Suppose we test m hypotheses, m_0 of which are truly null. Let V denote the number of type I errors, and R the total number of rejections.

$$FWER = P(V > 0) \qquad FDR = \mathbb{E}[FDP]$$

① If $m_0 = m$, then FWER = FDR

Pf: Since $m_0 = m$, either $R = 0$, or $R > 0$ and $V = R$

$$\Rightarrow FDP = \begin{cases} 1 & R > 0 \\ 0 & R = 0 \end{cases}$$

$$\mathbb{E}[FDP] = 1 \cdot P(R > 0) + 0 \cdot P(R = 0) = P(R > 0)$$
$$= P(V > 0) = FWER \quad //$$

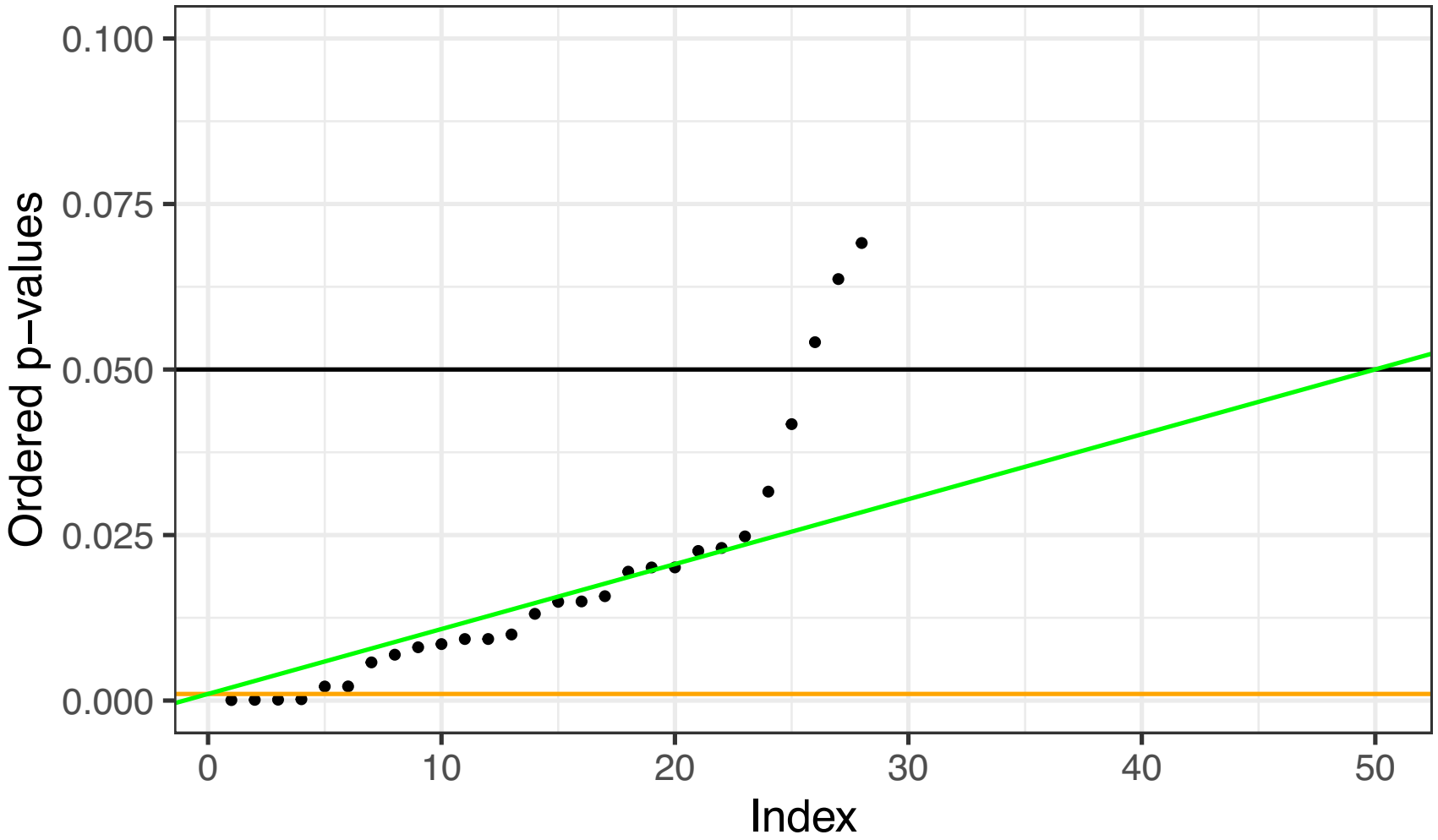
② In general, FDR $\leq$ FWER

Pf: $FDP \leq \mathbb{1}\{V > 0\}$

$$\Rightarrow FDR = \mathbb{E}[FDP] \leq \mathbb{E}[\mathbb{1}\{V > 0\}]$$
$$= P(V > 0) = FWER \quad //$$

The Benjamini-Hochberg procedure

- Order the p-values $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}$
- Let $i^* = \max\left\{i : p_{(i)} < \frac{i\alpha}{m}\right\}$
- Reject $H_{0,(i)}$ for all $i \leq i^*$



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- Let $i^* = \max\left\{i : p_{(i)} < \frac{i\alpha}{m}\right\}$
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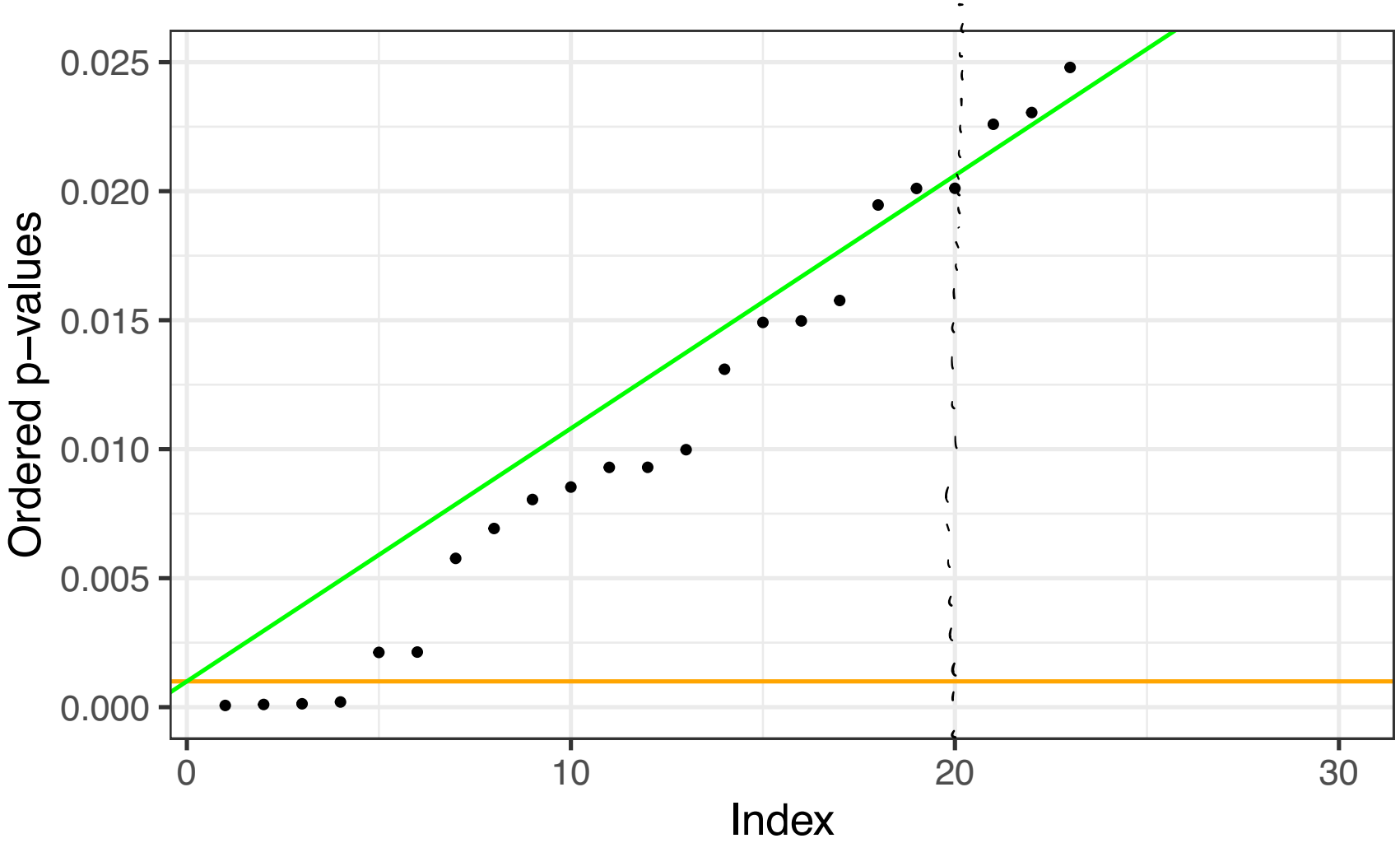
reject for $p_{(1)}, \ldots, p_{(20)}$

fail to reject the rest



The Benjamini-Hochberg procedure: intuition

<u>Claim</u>: If the hypotheses are independent, BH controls FDR at level $\frac{m_0}{m}\alpha \leq \alpha$

<u>Intuition</u>: If H_0 is true, $P(p_i < t) \leq t$

(in general, under H_0 we expect p-values to be $U(0,1)$)

$i^* = \max\{i : p_{(i)} < \frac{i\alpha}{m}\}$ $\Rightarrow$ reject all p-values p_j st $p_j < \frac{i^*\alpha}{m}$

If H_0 is true, $P(p_j < \frac{i^*\alpha}{m}) = \frac{i^*\alpha}{m}$ $\quad (p_j \sim U(0,1))$

we have m_0 true null hypotheses

$\Rightarrow$ we expect $m_0 \cdot \frac{i^*\alpha}{m}$ type I errors

$$\Rightarrow \quad FDP \approx \frac{m_0 \cdot \frac{i^*\alpha}{m}}{i^*} \qquad i^* = \#\text{ rejections total}$$

$$= \frac{m_0}{m}\alpha \leq \alpha$$